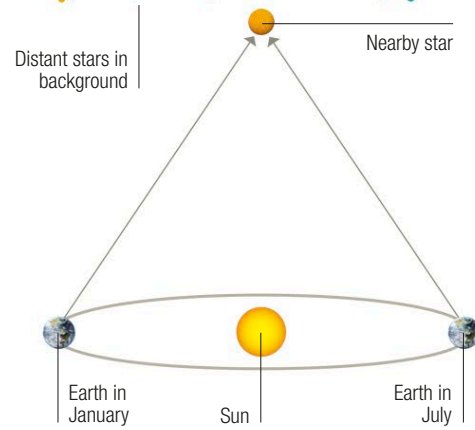


Distance

Stars other than the Sun are so far away that a special unit is needed to express the distance to them. This unit is the light-year and is the distance light travels through space in a year, which is about 5.9 trillion miles (9.5 trillion km). The 100 brightest stars we can see in the night sky vary from 4.4 to around 2,500 light-years away. The distances to stars can be measured in various ways. For relatively nearby stars, a method called parallax is used (see right). For more remote stars, astronomers have to use more complex indirect methods. Because these methods are less precise, the distances to many stars, even to some of the brightest in the sky, are known only approximately.



◁ Parallax method
If a nearby star is viewed from Earth on two occasions, when Earth is at opposite sides of its orbit around the Sun, the nearby star seems to shift a little against the background of more distant stars. The amount of shift provides the basis for calculating how far away the star is.

